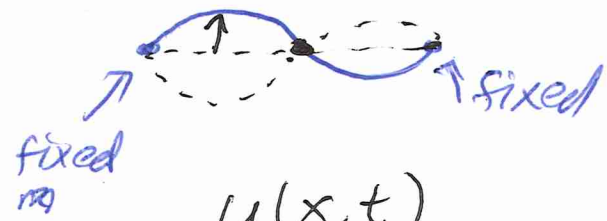
ch 6 §4 Wave eq'n

Vibration of a guitar string



$u(x,t)$

u is the displacement from equil.

Assumptions

string vibrates in a plane (true!)
small deformations
density is constant.

$$u_{xx} - \frac{1}{c^2} u_{tt} = 0$$

c is a constant (a fn of elasticity constant, density)

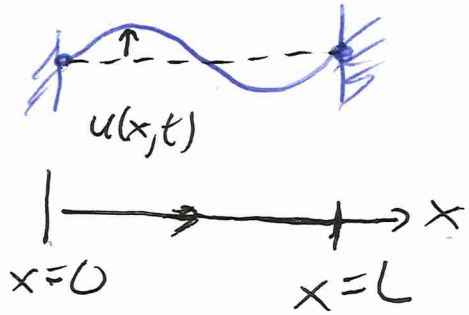
$$t' = ct$$

$$\frac{\partial u}{\partial t} = \frac{\partial u}{\partial t'} \frac{dt'}{dt} = c\, u_{t'}$$

$$\frac{\partial^2 u}{\partial t} = \frac{\partial}{\partial t}\left(\frac{\partial u}{\partial t}\right) = \frac{\partial}{\partial t}(c\, u_{t'}) = c \frac{\partial u_{t'}}{\partial t'} \frac{dt'}{dt} = c^2 u_{t't'}$$

$$u_{xx} - \frac{1}{c^2}(c^2 u_{t't'}) = 0 \qquad u_{xx} - u_{tt} = 0$$

$u_{xx} - u_{tt} = 0 \qquad 0 < x < L \qquad t > 0$



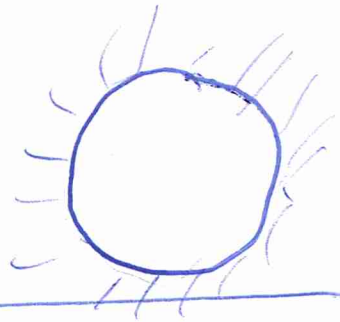
I.C. $u(x,0) = \phi(x) \quad 0 < x < L$ (initial displacement)

$u_t(x,0) = \psi(x) \quad 0 < x < L$ (velocity)

B.C. $u(0,t) = f(t), \quad u(L,t) = g(t) \quad t \geq 0$

Sol'n: Derived in §5.7

<u>d'Alembert sol'n</u>



$u_{xx} - u_{yy} = 0$

$\xi = x + t \qquad \eta = x - t$

$u_{\xi\eta} = 0$

$u_\xi = f(\xi)$

$u = F(\xi) + G(\eta)$

$$\boxed{u(x,t) = F(x+t) + G(x-t)}$$

Use I.C. to get (HW)

$$u(x,t) = \frac{1}{2}\left[\phi(x+t) + \phi(x-t)\right] + \frac{1}{2}\int_{x-t}^{x+t} \psi(\tau)\, d\tau$$

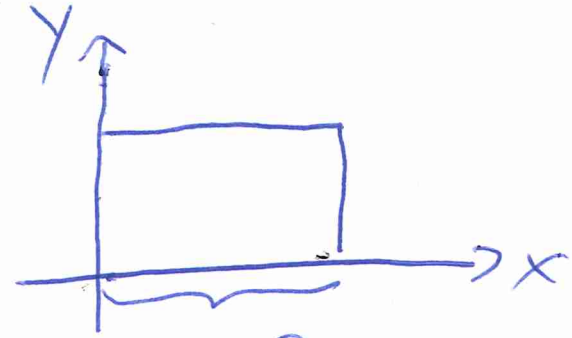
(You will need Leibniz rule to show that $u(x,t)$ is a sol'n)

Example ~~of~~ where ③ is violated

$\Delta u = 0$ on Ω

$u(x,0) = 0 \quad 0 < x < L$

$\frac{\partial u}{\partial y}(x,0) = \underbrace{\frac{1}{n}\sin(nx)}_{0}$



$u = 0$

$\frac{\partial u}{\partial y} = \frac{1}{n}\sin(nx)$

$u(x,y) = 0$

The sol'n: $u(x,y) = \frac{1}{n^2}\sinh(ny)\sin(nx)$

$\sinh(ny) = \frac{1}{2}(e^{ny} - e^{-ny})$

$\lim_{n\to\infty} \frac{1}{n}\sin(nx) = 0$

$\lim_{n\to\infty} \frac{e^{ny}}{n^2} = \infty$

$\lim_{n\to\infty} u(x,y) \to \infty$

§6.5 Well-Posed Problem

Def'n A problem in PDE is <u>well-posed</u> if

1. A sol'n exists
2. The sol'n is unique
3. The sol'n depends continuously on data

About 3: We want small changes in data (IC, BC, coeff) to produce only small changes in sol'n.

~~example where~~ More on ③ ~~is violated~~.

Laplace eqn $\Delta u = 0$ $(u_{xx} + u_{yy} = 0)$

suppose u is a sol'n of

$$\Delta u = 0 \text{ in } \Omega$$
$$u(\underline{x}) = f \text{ on } \partial\Omega$$

only data is f.

small changes in $f \Rightarrow$ small changes in u.

suppose $\tilde{u}$ is a sol'n of

$$\Delta \tilde{u} = 0 \text{ in } \Omega$$
$$\tilde{u}(\underline{x}) = \tilde{f} \text{ on } \partial\Omega$$

where $|f(x,y) - \tilde{f}(x,y)| < \delta$ (small)

If well-posed $\max_{(x,y)\in\bar{\Omega}} |u(x,y) - \tilde{u}(x,y)| < \varepsilon(\delta)$

$\Rightarrow$ sol'n depends continuously on data.

Ch 7 Laplace Eq'n

$$\Delta u = \frac{\partial^2 u}{\partial x_1^2} + \frac{\partial^2 u}{\partial x_2^2} + \dots + \frac{\partial^2 u}{\partial x_n^2} = 0$$

<u>Def'n</u> A f'n, u, is called <u>harmonic</u> in the domain Ω, if $u \in C^2(\Omega)$ and satisfies $\Delta u = 0$ in Ω.

<u>Thm 1.1</u> If $u \in C^0(\Omega)$ and $\Delta u = 0$ Then u is analytic (i.e. There ~~exit~~ ~~exit~~ exists a Taylor series expansion that equals the f'n)

Recall: analytic f'ns $\subset C^\infty$

$\boxed{\text{① harmonic f'ns are } C^\infty \text{ (very smooth)}}$

Note: There are $u \notin C^2$ which satisfy $\Delta u = 0$. According to Thm, what does this imply about u? $\Rightarrow u \notin C^0$

ex) $\dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 u}{\partial y^2} = 0$ $\qquad z = x + iy$

$$u(x,y) = \begin{cases} \operatorname{Re}\left(e^{-\frac{1}{z^4}}\right) & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$$

$\Rightarrow$ All 1st + 2nd partials exist, even at $(0,0)$

but $u \notin C^0(\mathbb{R}^2)$

Laplace eq'n in polar coord

$x = r\cos\theta \qquad y = r\sin\theta$

$x^2 + y^2 = r^2 \qquad \tan\theta = y/x$

For $n=2$ $\quad \Delta u = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial u}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 u}{\partial \theta^2} = 0$ (*)

(*) See reference, strauss, for details on ~~getting~~ change of variables.